

LINUX PERMISSION USING SYMBOLIC & NUMERIC METHODS ASSIGNMENT

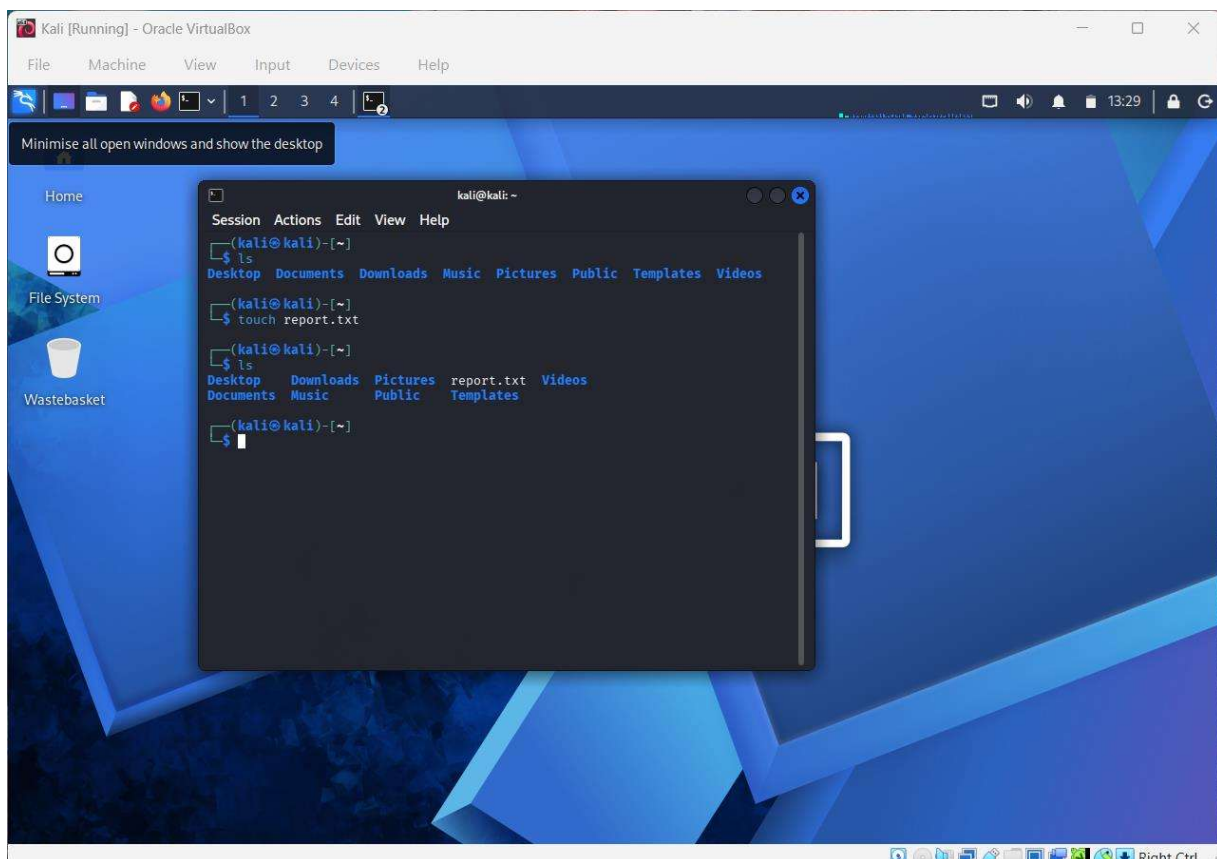
Section 1: File Permission Assignment

Scenario

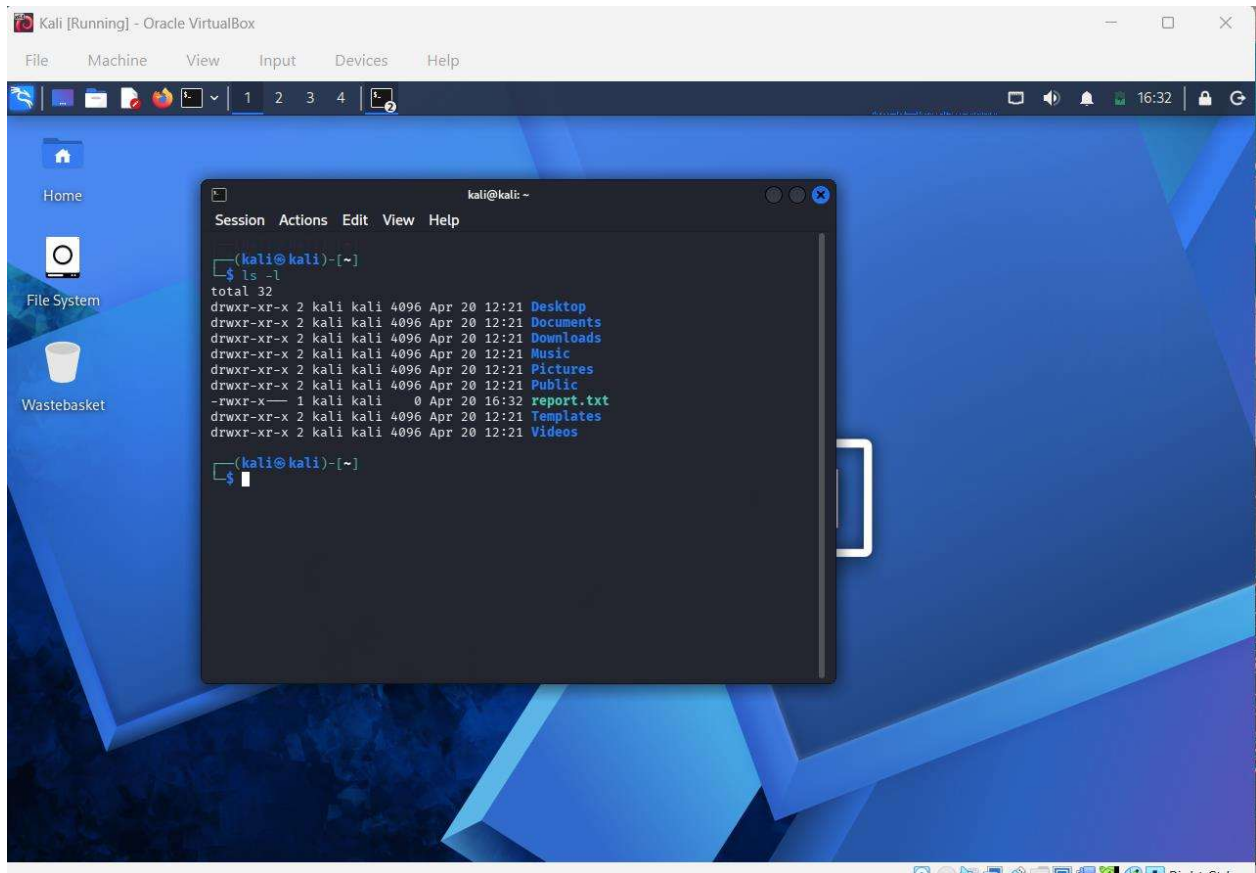
You are the Linux administrator for a small team. A file named **report.txt** has been created in a shared working directory. Your task is to inspect and change permissions on the file using both the symbolic and numeric methods.

Task:

- Create a file named report.txt.
- Verify that the file exists.



Check the current permissions on the file.



The screenshot shows a Kali Linux desktop environment within an Oracle VM VirtualBox window. The desktop has a blue geometric background and icons for Home, File System, and Wastebasket. A terminal window is open, displaying the command `ls -l` and its output. The output lists the permissions, owner, group, size, date, and name for the current directory's contents.

```
kali@kali: ~  
$ ls -l  
total 32  
drwxr-xr-x 2 kali kali 4096 Apr 20 12:21 Desktop  
drwxr-xr-x 2 kali kali 4096 Apr 20 12:21 Documents  
drwxr-xr-x 2 kali kali 4096 Apr 20 12:21 Downloads  
drwxr-xr-x 2 kali kali 4096 Apr 20 12:21 Music  
drwxr-xr-x 2 kali kali 4096 Apr 20 12:21 Pictures  
drwxr-xr-x 2 kali kali 4096 Apr 20 12:21 Public  
-rwxr-x--- 1 kali kali 0 Apr 20 16:32 report.txt  
drwxr-xr-x 2 kali kali 4096 Apr 20 12:21 Templates  
drwxr-xr-x 2 kali kali 4096 Apr 20 12:21 Videos
```

Using the symbolic method only, set the permissions so that the owner has read and write, the group has read only, and others have no permission.

Verify the result and record the final permission string.

```

kali@kali ~
Session Actions Edit View Help
drwxr-xr-x 2 kali kali 4096 Apr 20 12:21 Music
drwxr-xr-x 2 kali kali 4096 Apr 20 12:21 Pictures
drwxr-xr-x 2 kali kali 4096 Apr 20 12:21 Public
-rw-r--r-- 1 kali kali 0 Apr 20 13:29 report.txt
drwxr-xr-x 2 kali kali 4096 Apr 20 12:21 Templates
drwxr-xr-x 2 kali kali 4096 Apr 20 12:21 Videos

(kali@kali)~
$ chmod g-w report.txt

(kali@kali)~
$ chmod o-r report.txt

(kali@kali)~
$ ls -l
total 32
drwxr-xr-x 2 kali kali 4096 Apr 20 12:21 Desktop
drwxr-xr-x 2 kali kali 4096 Apr 20 12:21 Documents
drwxr-xr-x 2 kali kali 4096 Apr 20 12:21 Downloads
drwxr-xr-x 2 kali kali 4096 Apr 20 12:21 Music
drwxr-xr-x 2 kali kali 4096 Apr 20 12:21 Pictures
drwxr-xr-x 2 kali kali 4096 Apr 20 12:21 Public
-rw-r--r-- 1 kali kali 0 Apr 20 13:29 report.txt
drwxr-xr-x 2 kali kali 4096 Apr 20 12:21 Templates
drwxr-xr-x 2 kali kali 4096 Apr 20 12:21 Videos

(kali@kali)~
$
  
```

Using the numeric method only, change the file permissions so that the owner has read, write, and execute, the group has read and execute, and others have no permission.

Verify the result and record the final permission string.

```

kali@kali ~
Session Actions Edit View Help
drwxr-xr-x 2 kali kali 4096 Apr 20 12:21 Public
-rw-r----- 1 kali kali 0 Apr 20 13:29 report.txt
drwxr-xr-x 2 kali kali 4096 Apr 20 12:21 Templates
drwxr-xr-x 2 kali kali 4096 Apr 20 12:21 Videos

(kali@kali)~$ chmod 750 report.txt

(kali@kali)~$ ls
Desktop Downloads Pictures report.txt Videos
Documents Music Public Templates

(kali@kali)~$ ls -l
total 32
drwxr-xr-x 2 kali kali 4096 Apr 20 12:21 Desktop
drwxr-xr-x 2 kali kali 4096 Apr 20 12:21 Documents
drwxr-xr-x 2 kali kali 4096 Apr 20 12:21 Downloads
drwxr-xr-x 2 kali kali 4096 Apr 20 12:21 Music
drwxr-xr-x 2 kali kali 4096 Apr 20 12:21 Pictures
drwxr-xr-x 2 kali kali 4096 Apr 20 12:21 Public
-rw-r-xr-x 1 kali kali 0 Apr 20 13:29 report.txt
drwxr-xr-x 2 kali kali 4096 Apr 20 12:21 Templates
drwxr-xr-x 2 kali kali 4096 Apr 20 12:21 Videos

(kali@kali)~$
  
```

- After the symbolic task, the permission string is rw-r-----.
- After the numeric task, the permission string should be rwxr-xr-x.

Reflection Questions

1. Symbolic vs. Numeric Methods

The Symbolic Method

This method uses letters to represent who you are changing permissions for and what action you are taking. It's very intuitive because it reads like a sentence.

- **Who:** u (user/owner), g (group), o (others), a (all).
- **Action:** + (add), - (remove), = (set exactly).
- **Permission:** r (read), w (write), x (execute).

The Numeric (Octal) Method

This method uses a three-digit code where each digit represents a specific level of access for the Owner, Group, and Others. Each permission is assigned a mathematical value:

- **4:** Read (r)
- **2:** Write (w)
- **1:** Execute (x)
- **0:** No permission

2. What does 640 means?

In the numeric method, **640** translates to:

- **6 (Owner):** Read (4) + Write (2). The owner can edit the file.
- **4 (Group):** Read (4) only. The group can see it but not change it.
- **0 (Others):** No permissions. Everyone else is locked out.

Summary: The owner can read/write, the group can only read, and others have no access.

3. What does 750 mean?

In the numeric method, **750** translates to:

- **7 (Owner):** Read (4) + Write (2) + Execute (1). Full control.
- **5 (Group):** Read (4) + Execute (1). They can view the file or run it (if it's a script/folder).
- **0 (Others):** No permissions.

The Numeric Method is almost always faster because:

1. **Speed of Input:** Typing `chmod 755` is significantly faster than typing `chmod u=rwx,go=rx`.
2. **Total State Reset:** The numeric method sets the *exact* state of all permissions at once. If a file has messy, unknown permissions, a numeric command "wipes the slate clean" and applies to the exact profile you want.
3. **Automation:** When writing scripts or deployment manifests, using numbers is the industry standard because it is concise and avoids the ambiguity of adding/subtracting bits.

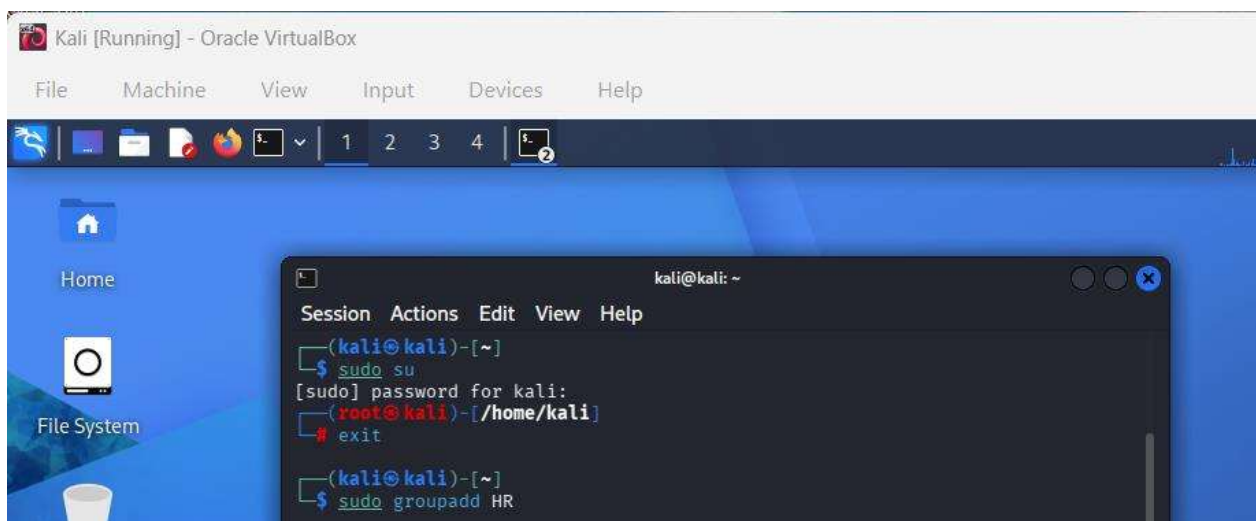
Section 2: User and Group Administration

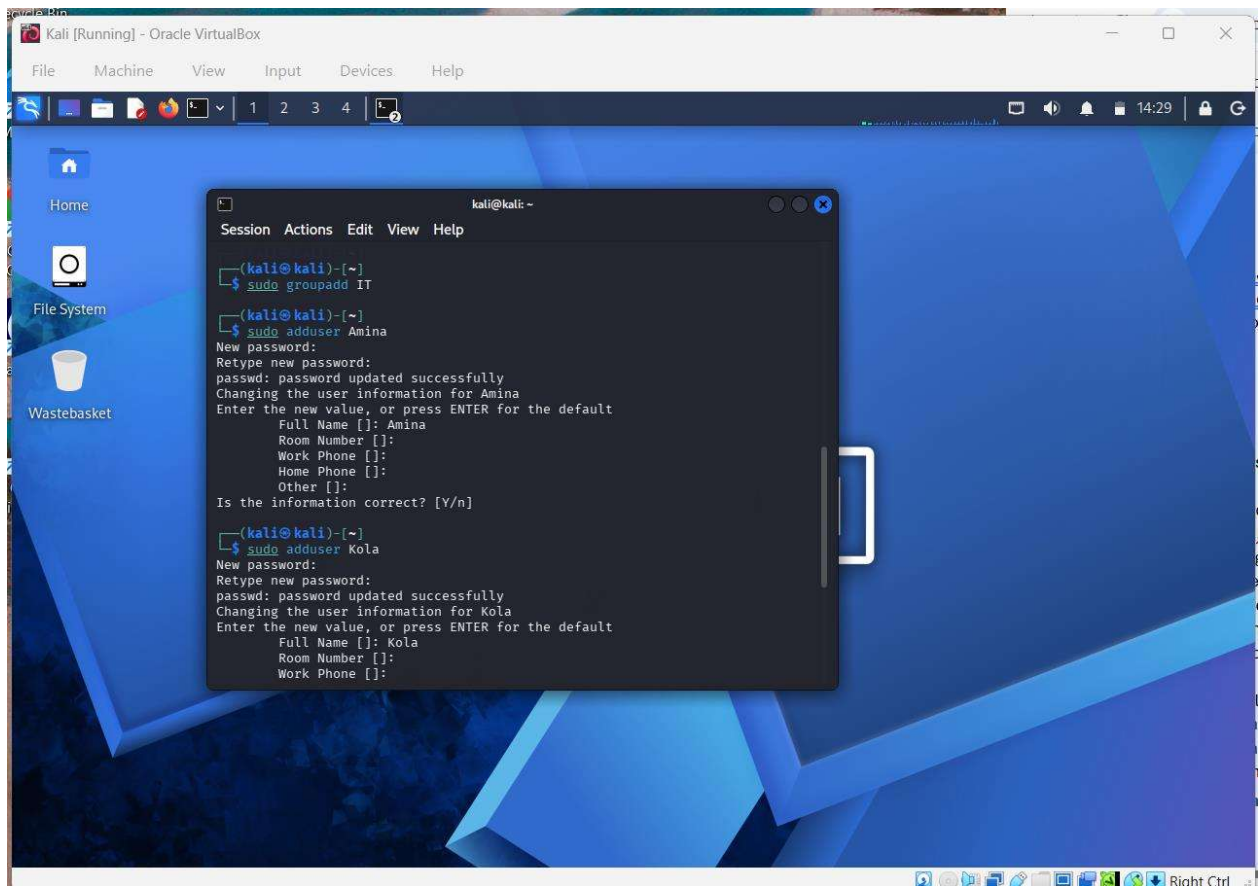
Scenario

You have been asked to prepare a Linux system for two departments: **HR** and **IT**. You must create the required groups, create users, add the users to their correct groups, then modify memberships and file access as directed.

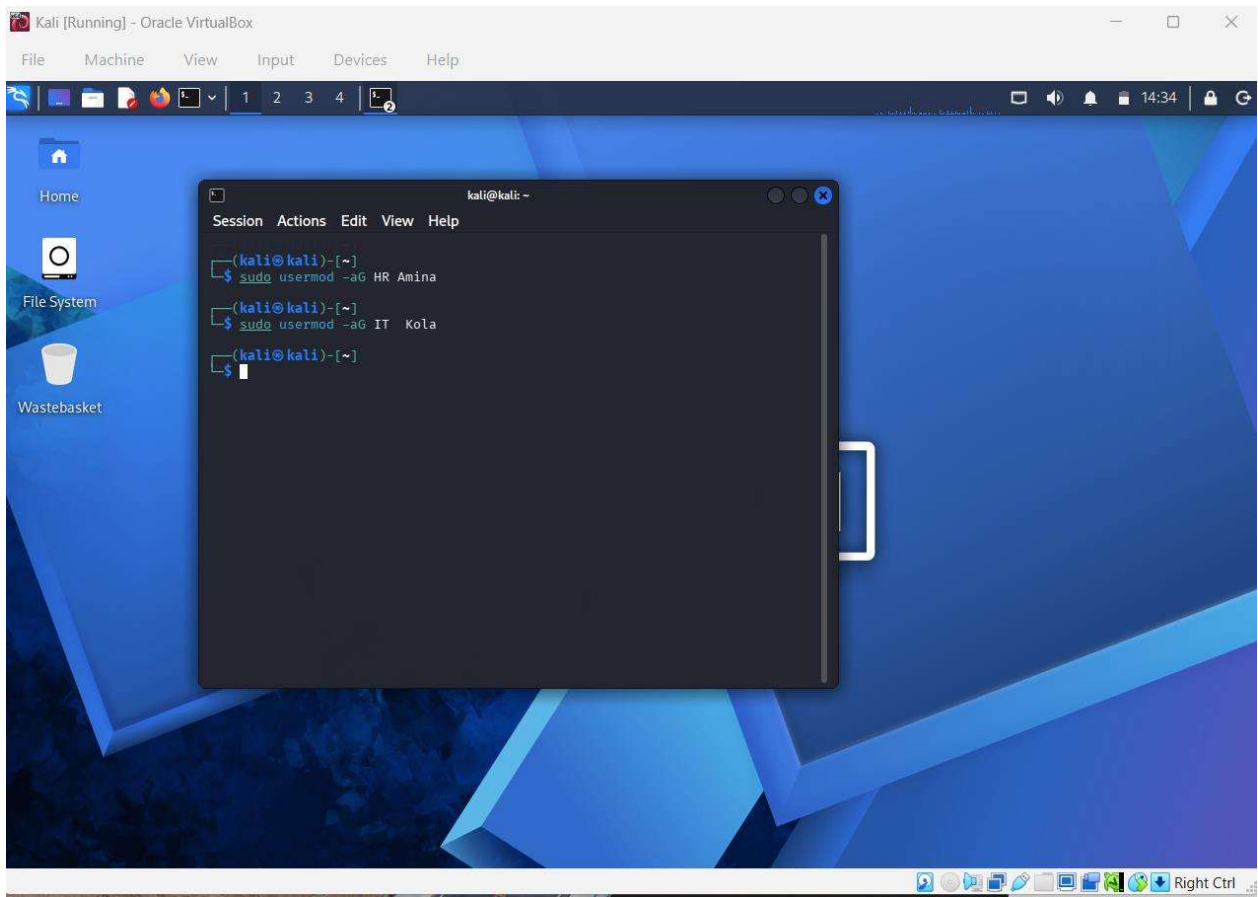
Tasks: Create two groups named HR and IT.

- Create two users named amina and kola.
- Assign a password to each user.

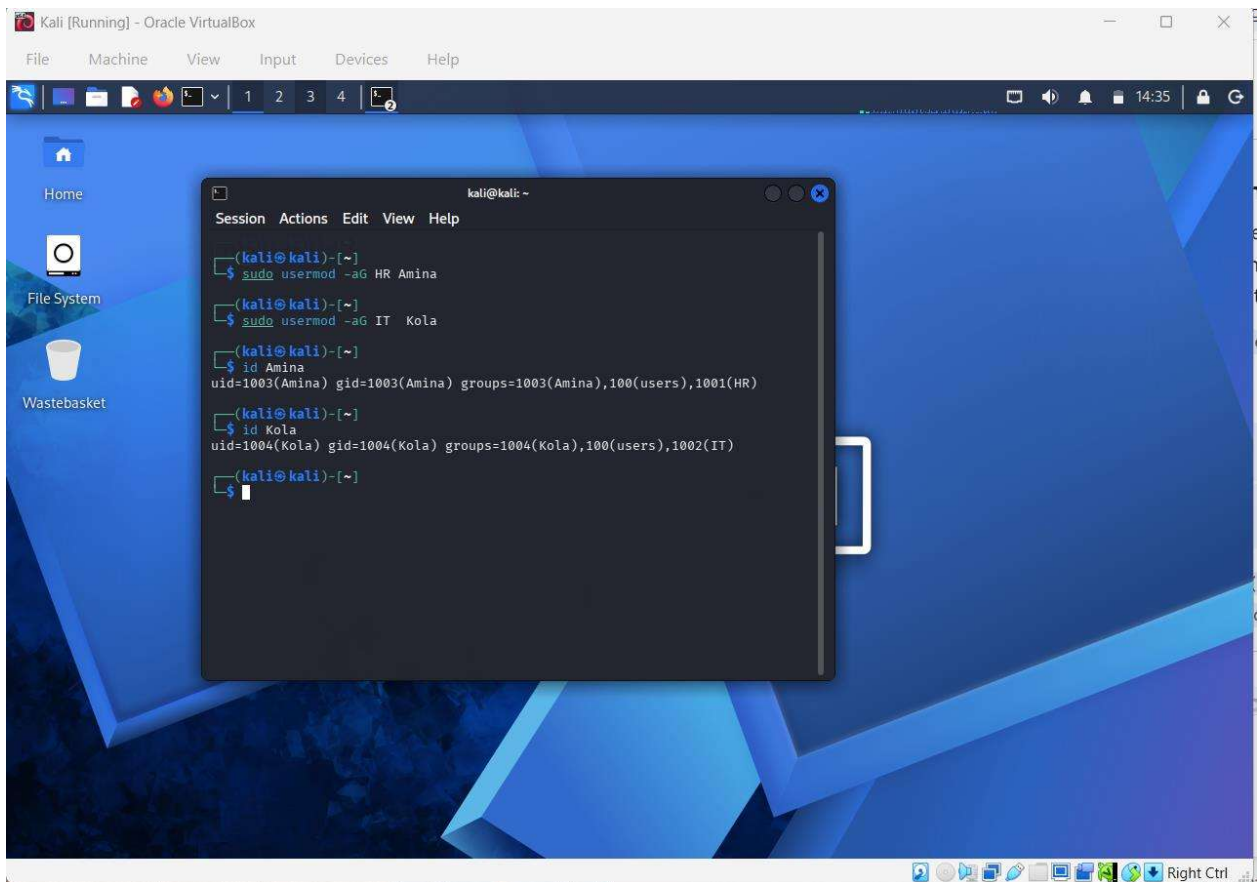




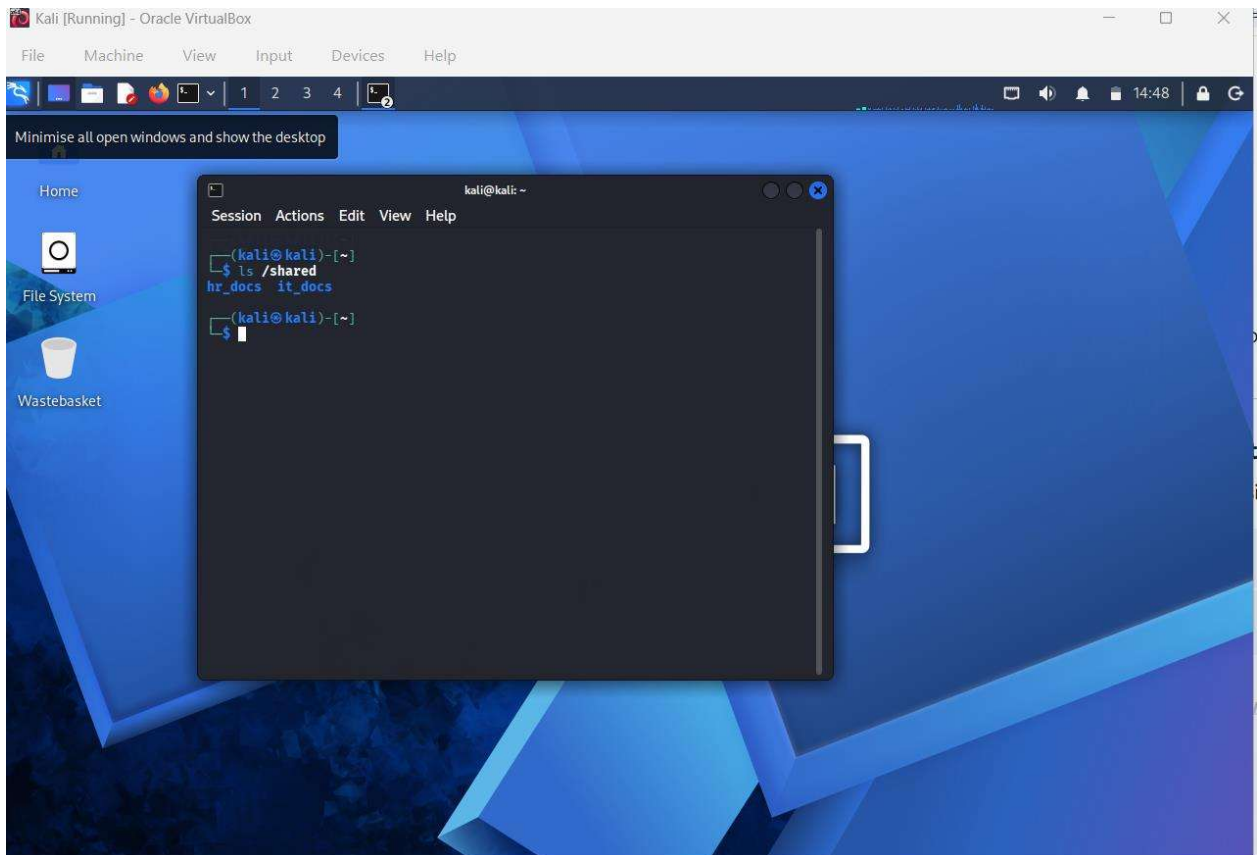
- Add amina to the hr group.
- Add kola to the it group.



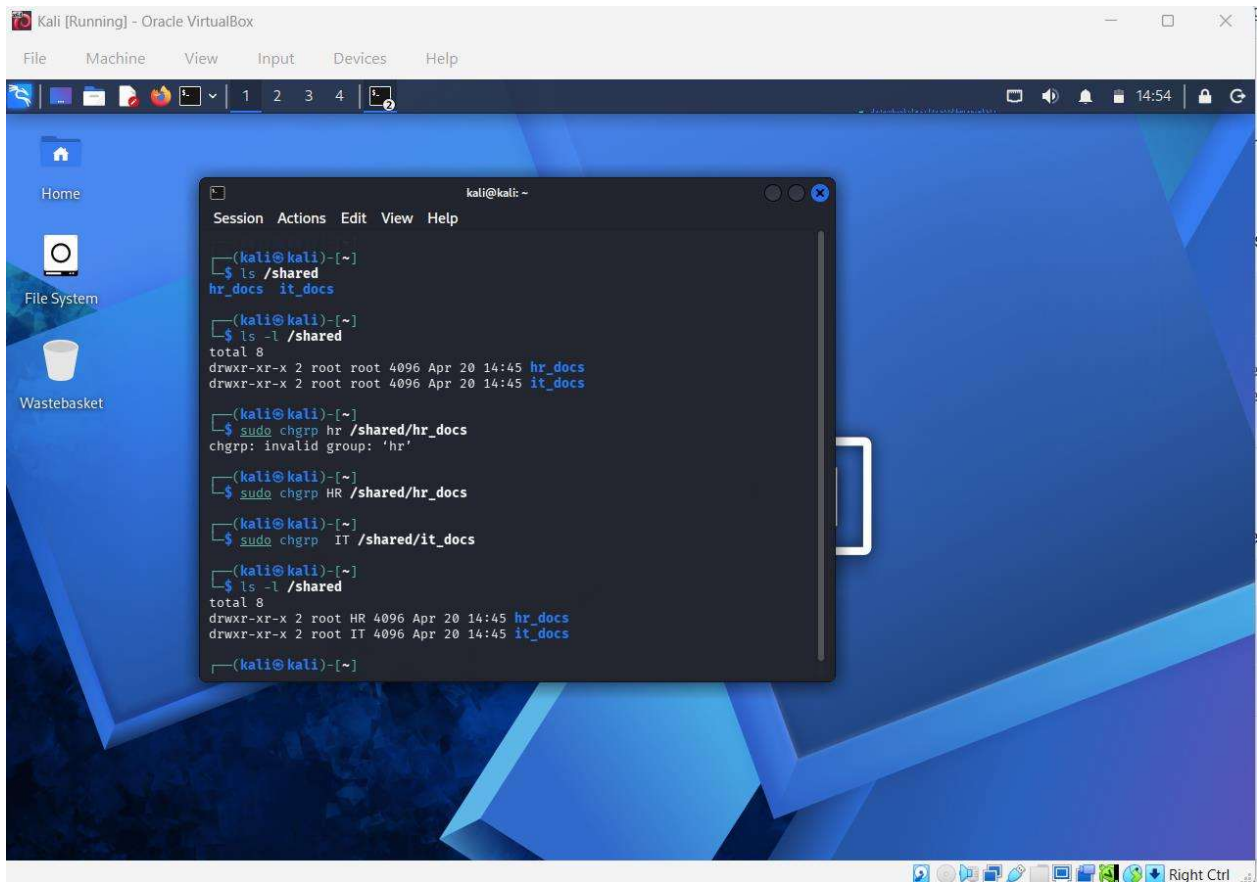
- Confirm each user's group membership using the appropriate verification command.



- Create two directories named `/shared/hr_docs` and `/shared/it_docs`.

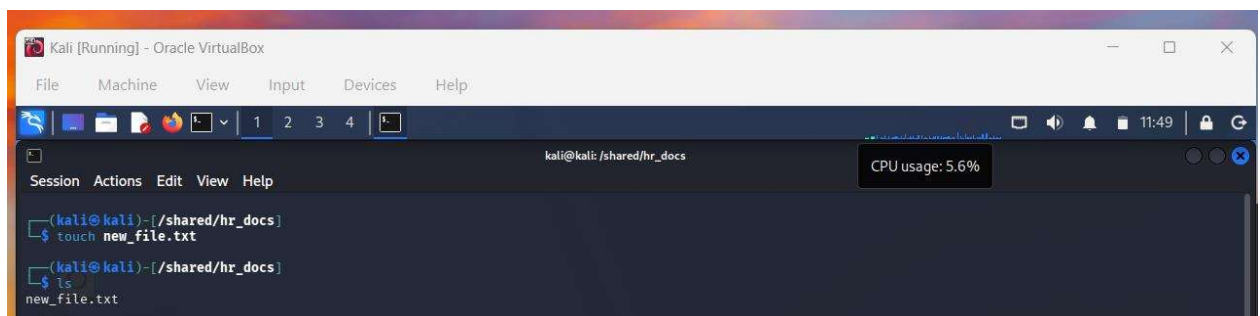
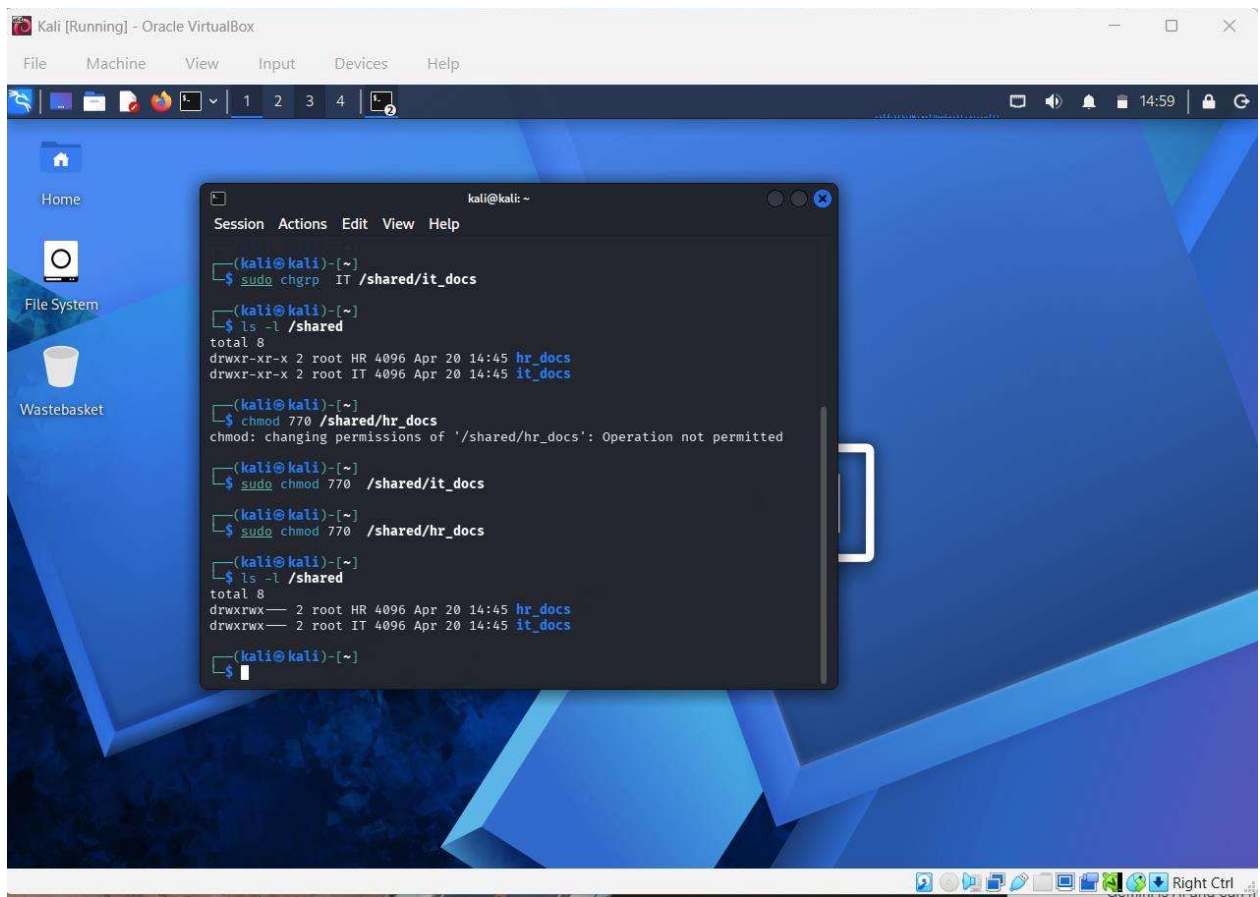


- Set the group ownership of hr_docs to hr and the group ownership of it_docs to it.



```
kali@kali ~  
Session Actions Edit View Help  
[kali@kali]~  
$ ls /shared  
hr_docs it_docs  
[kali@kali]~  
$ ls -l /shared  
total 8  
drwxr-xr-x 2 root root 4096 Apr 20 14:45 hr_docs  
drwxr-xr-x 2 root root 4096 Apr 20 14:45 it_docs  
[kali@kali]~  
$ sudo chgrp hr /shared/hr_docs  
chgrp: invalid group: 'hr'  
[kali@kali]~  
$ sudo chgrp HR /shared/hr_docs  
[kali@kali]~  
$ sudo chgrp IT /shared/it_docs  
[kali@kali]~  
$ ls -l /shared  
total 8  
drwxr-xr-x 2 root HR 4096 Apr 20 14:45 hr_docs  
drwxr-xr-x 2 root IT 4096 Apr 20 14:45 it_docs  
[kali@kali]~
```

- Modify the directory permissions so that only the owner and the assigned group can access each directory.



```

kali@kali: /shared/it_docs

Session Actions Edit View Help
(kali@kali)-[/home]
$ cd /shared/it_docs
cd: permission denied: /shared/it_docs

(kali@kali)-[/home]
$ cd /shared
(kali@kali)-[/shared]
$ ls
it_docs
(kali@kali)-[/shared]
$ cd /shared/it_docs
cd: permission denied: /shared/it_docs

(kali@kali)-[/shared]
$ sudo /shared/it_docs
[sudo] password for kali:
sudo: /shared/it_docs: command not found

(kali@kali)-[/shared]
$ sudo chown kali: kali it_docs
chown: cannot access 'kali': No such file or directory

(kali@kali)-[/shared]
$ sudo chown kali:kali it_docs
(kali@kali)-[/shared]
$ ls
it_docs
(kali@kali)-[/shared]
$ cd it_docs
(kali@kali)-[/shared/it_docs]
$ touch sample.log
(kali@kali)-[/shared/it_docs]
$ ls
sample.log
(kali@kali)-[/shared/it_docs]
$

```

- Modify one user account by adding that user to an additional group.

```

kali@kali: ~
Session Actions Edit View Help
drwxr-xr-x 2 root HR 4096 Apr 20 14:45 hr_docs
drwxr-xr-x 2 root IT 4096 Apr 20 14:45 it_docs

(kali@kali)-[~]
$ chmod 770 /shared/hr_docs
chmod: changing permissions of '/shared/hr_docs': Operation not permitted

(kali@kali)-[~]
$ sudo chmod 770 /shared/it_docs

(kali@kali)-[~]
$ sudo chmod 770 /shared/hr_docs

(kali@kali)-[~]
$ ls -l /shared
total 8
drwxrwx--- 2 root HR 4096 Apr 20 14:45 hr_docs
drwxrwx--- 2 root IT 4096 Apr 20 14:45 it_docs

(kali@kali)-[~]
$ sudo usermod -aG IT Amina

(kali@kali)-[~]
$ groups Amina
Amina : Amina users HR IT

(kali@kali)-[~]
$
  
```

- Modify one user account by removing the user from a group.
- Ensure that HR users can access hr_docs but not it_docs.
- Ensure that IT users can access it_docs but not hr_docs.
- Verify your access-control settings with command output.

Directory	Group Owner	Permissions	IT User Result
/shared/it_docs	it	rw-rwx---	Access Granted (Group Match)
/shared/hr_docs	hr	rw-rwx---	Access Denied (Permission 0)

Directory	Group Owner	Permissions	HR User Access
/shared/hr_docs	hr	rw-rwx---	ALLOWED (Matches Group)
/shared/it_docs	it	rw-rwx---	DENIED (User is "Other")


```
kali@kali ~  
Session Actions Edit View Help  
kali@kali ~  
$ sudo -u Amina ls /shared/it_docs  
ls: cannot open directory '/shared/it_docs': Permission denied  
kali@kali ~  
$ sudo -u Amina ls /shared/hr_docs  
[sudo] password for kali:  
kali@kali ~  
$ sudo -u Kola ls /shared/hr_docs  
ls: cannot open directory '/shared/hr_docs': Permission denied  
kali@kali ~  
$ sudo -u Kola ls /shared/it_docs  
ls: cannot open directory '/shared/it_docs': Permission denied  
kali@kali ~  
$
```

```

kali@kali: ~
Session Actions Edit View Help
~$ chmod 770 /shared/hr_docs
chmod: changing permissions of '/shared/hr_docs': Operation not permitted

kali@kali:~$ sudo chmod 770 /shared/it_docs

kali@kali:~$ sudo chmod 770 /shared/hr_docs

kali@kali:~$ ls -l /shared
total 8
drwxrwx--- 2 root HR 4096 Apr 20 14:45 hr_docs
drwxrwx--- 2 root IT 4096 Apr 20 14:45 it_docs

kali@kali:~$ sudo usermod -aG IT Amina

kali@kali:~$ groups Amina
Amina : Amina users HR IT

kali@kali:~$ sudo gpasswd -d Amina IT
Removing user Amina from group IT

kali@kali:~$
  
```

```

kali@kali: ~
Session Actions Edit View Help
drwxr-xr-x 2 root HR 4096 Apr 20 14:45 hr_docs
drwxr-xr-x 2 root IT 4096 Apr 20 14:45 it_docs

(kali@kali)-[~]
$ chmod 770 /shared/hr_docs
chmod: changing permissions of '/shared/hr_docs': Operation not permitted

(kali@kali)-[~]
$ sudo chmod 770 /shared/it_docs

(kali@kali)-[~]
$ sudo chmod 770 /shared/hr_docs

(kali@kali)-[~]
$ ls -l /shared
total 8
drwxrwx--- 2 root HR 4096 Apr 20 14:45 hr_docs
drwxrwx--- 2 root IT 4096 Apr 20 14:45 it_docs

(kali@kali)-[~]
$ sudo usermod -aG IT Amina

(kali@kali)-[~]
$ groups Amina
Amina : Amina users HR IT

(kali@kali)-[~]
$

```

Linux evaluates permissions in a specific order: **User** → **Group** → **Others**.

- Every file/directory has a **Group Owner**.
- Every group has a set of permissions (like rwx for Read, Write, and Execute).

If you are added to a group (like **hr**), the system suddenly recognizes you as a member of that group, Instead of falling into the restricted "**Others**" category, you are upgraded to the "**Group**" category and inherit all the permissions assigned to it.

As you saw with your **it_docs** folder, directories require **Execute (x)** permission to enter.

- If a directory is set to **drwxrwx---** (Owner: root, Group: it), a normal user is blocked.
- The moment you become a member of the **it** group, the system grants you that middle **rwx** set, allowing you to **cd** into the folder and view its contents.